Learning how to prove Workshop Week 11

Factor groups

$$\begin{tikzcd} G \arrow[r, "f"] \arrow[d, "\pi"'] & H \\ G/\ker(f) \arrow[ur, dashed, "\tilde{f}"'] & \end{tikzcd}$$

Normal subgroups. Factor groups

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1. Let H be a normal subgroup in a group G such that $|H| = 2$. Show that $H \subseteq Z(G)$.

> **Solution:** Write $H = \{1, h\}$ for some $h \in G$, $h \neq 1$. To show that $H \subseteq Z(G)$ we only need to prove that $hg = gh$ for any $g \in G$. But given $g \in G$ since $H \lhd G$ we have that $gHg^{-1} = H$. We have two possibilities either $ghg^{-1} = 1$ or $ghg^{-1} = h$. If $ghg^{-1} = 1$, then $h = g^{-1}g = 1$, a contradiction. Thus, $ghg^{-1} = h$ and $gh = hg$ and we are done.

2. Let H be a subgroup of a group G such that $H \subseteq Z(G)$ and G/H is cyclic. Prove that G is abelian.

> **Solution:** Let us begin by noticing that H is a normal subgroup of G because $H \subseteq Z(G)$, so it makes sense to consider the factor group G/H.
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> Suppose that $G/H = \langle Hg \rangle$ for some $g \in G$. Given g_1, $g_2 \in G$ consider the right cosets Hg_1, Hg_2. Because G/H is cyclic, we can find $n, m \in \mathbb{Z}$ such that $Hg_1 = (Hg)^n = Hg^n$ and $Hg_2 = (Hg)^m = Hg^m$. From here, we obtain that $g_1 = h_1 g^n$ and $g_2 = h_2 g^m$ for some h_1, $h_2 \in H$. Therefore:
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> $$g_1 g_2 = (h_1 g^n)(h_2 g^m) = h_1 h_2 g^{n+m} = g_2 g_1,$$
>
> because $H \subseteq Z(G)$, by hypothesis. This proves that G is abelian.